Data & Methods draft: Fernhill Software, version 1 of 3 (the enthusiast's model)

1. Unit of observation

One row of my data is one product team in one quarter, because each team has its own manager and delivery is counted by the quarter. There are 1,440 rows, from 120 teams over 12 quarters, July 2023 to June 2026, eight before the reorganization and four after. All 120 teams changed managers in July 2025, and AI licenses arrived the same month, so `manager_type`, `period` and `ai_access` move together and make one comparison, not three. Each team has its own manager, as the Brief says (the file has no manager ID to check), so a row is also one manager's quarter, which is what my model is about.

2. Variable table

Concept	Variable	Kind	Source	What it misses
Manager type	`manager_type`	Labels	HR system	Skill within a type
Before or after	`period`	Labels	Calendar	Half-done quarters
People skill	`people_rating`	Measurement (share)	Upward survey	Likability and non-response
AI	`ai_access`	Yes/No	IT licenses	Actual use
Difficulty	`hard_close`	Yes/No	Close review	Difficulty seen beforehand
Delivery	`delivered_share`	Measurement (share)	Planning tool	Customer value and renegotiation

3. Data source

Fernhill's HR system, quarterly upward feedback survey, IT licenses, quarter-close review, and planning tool, read through People Analytics, cover all 120 teams for all 12 quarters. The file also holds records my model doesn't call for (team size, escalations, waits, departures). Gemini suggested describing them, and I left them out. Fernhill is a useful fiction, so these records are simulated, but not from the model above. They come from a model this version doesn't see, whose units are these team-quarters and whose relationships, noise, and assumptions are published in Module 3. Until then, I treat them as the firm's records.

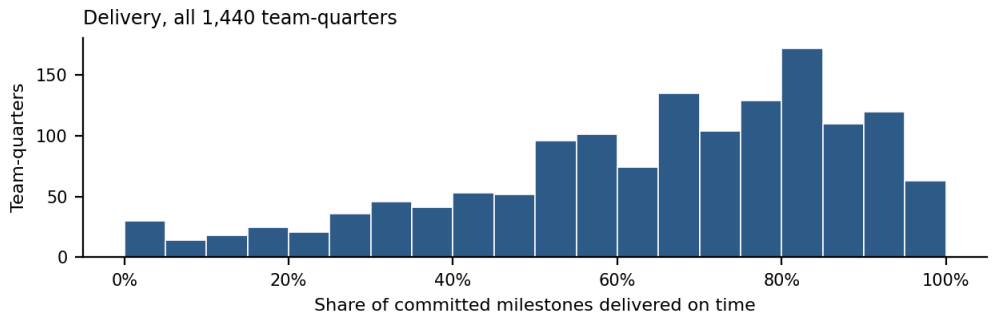
4. Quality concern

The people-skill rating is the most likely to mislead, being a survey answer and not a count. It is **missing** for 102 of 1,440 team-quarters (7%), and may be **mismeasured**, since favorable answers can reward a likable manager, and a manager new in July 2025 is rated after one quarter. If so, I would see blanks bunched in some teams or quarters, and new managers' ratings jumping in their first quarters. I will report the blanks and never fill them in, compare each team with itself before and after, and read the rating as the team's view, not a measure of skill. A team compared with itself cannot separate a new manager's first quarter from the manager's type, so I will say so.

5. First description

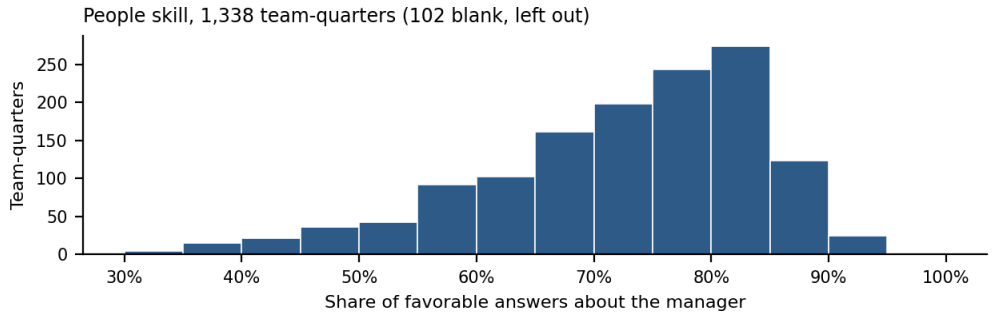
Delivery (`delivered_share`). *Kind:* a measurement, a share from 0 to 1. *How many:* 1,440, none blank. *Middle:* median 0.69, and mean 0.65, which a tail of bad quarters pulls down. *Spread:* half fall between 0.50

and 0.83, and the range is 0 to 1. *Shape:* skewed left, with a long tail of quarters that delivered little (30 under 5%). *Choice:* each team-quarter counts once, whatever its size. Weighting by milestones gives 0.64.



Against my model: not yet able to say. The model predicts delivery rises after July 2025, by about 5 points after a first-quarter dip, and one variable can't show whether it did. That comparison belongs in Results.

People skill (`people_rating`). *Kind:* a measurement, a share of favorable answers. *How many:* 1,338, with 102 blank (7%). The blank rate is the same before July 2025 as after (7.2% against 6.9%), and they are not bunched, since they come from 71 teams, at most 3 each. *Middle:* median 0.74, mean 0.72. *Spread:* half fall between 0.65 and 0.81, and the range is 0.31 to 0.94. *Shape:* skewed left, with a tail of low-rated team-quarters (77 below 0.50, spread over 23 teams). *Choice:* blanks left out, not filled in.



Against my model: not yet able to say. The model predicts higher ratings under professional managers, and one variable can't show that.

Difficulty (`hard_close`). *Kind:* yes/No. *How many:* 1,440, none blank. *Middle, spread and shape:* 0.43 rated hard (619 quarters). With two values, the proportion says it all. **Against my model:** not yet able to say. The model predicts a gain on hard quarters too, and 619 hard and 821 other quarters can test that.

Note on AI use

I used Gemini (Willamette account) for the counts, middles, spreads, and histograms. I added a milestone-weighted mean, and rejected its offer to fill blank ratings, since a filled-in rating is a measurement nobody took.